

A boundary counterexample to test-form density in $W_C^{1,2}$

Agent lane 14

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Source and question

The source paper is Andrea Carbonaro, Luca Tamanini, and Dario Trevisan, *Boundedness of Riesz transforms on $\text{RCD}(K, \infty)$ spaces*, arXiv:2308.16294. In Section 2, after defining test functions and test 1-forms by

$$\text{TestForm}(X) := \left\{ \sum_{i=1}^n g_i \, df_i : n \in \mathbb{N}, \, f_i, g_i \in \text{Test}(X) \right\},$$

the source recalls the associated covariant and Hodge Sobolev spaces $W_C^{1,2}(T^*X)$ and $W_H^{1,2}(T^*X)$. It then states that it is an open problem whether $\text{TestForm}(X)$ is dense in these Sobolev spaces, and denotes by $H_C^{1,2}(T^*X)$, $H_H^{1,2}(T^*X)$ the respective closures.

with the associated Sobolev spaces $W_C^{1,2}(T^*\mathbf{X})$, $W_H^{1,2}(T^*\mathbf{X})$, and norms

$$\begin{aligned} \|\omega\|_{W_C^{1,2}}^2 &:= \|\omega\|_{L^2(T^*\mathbf{X})}^2 + \|\nabla\omega\|_{L^2((T^*)^{\otimes 2}\mathbf{X})}^2, \\ \|\omega\|_{W_H^{1,2}}^2 &:= \|\omega\|_{L^2(T^*\mathbf{X})}^2 + \|\mathrm{d}\omega\|_{L^2(\Lambda^2 T^*\mathbf{X})}^2 + \|\mathrm{d}^*\omega\|_{L^2}^2 \end{aligned}$$

(see [22] for precise definitions, also of the spaces $L^2(T^{\otimes 2}\mathbf{X})$, $L^2(\Lambda^2 T^*\mathbf{X})$ and their “Hilbert-Schmidt” norms). It is an open problem to determine whether $\text{TestForm}(\mathbf{X})$ is dense in such Sobolev spaces: for this reason, in what follows we denote by $H_C^{1,2}(T^*\mathbf{X})$, $H_H^{1,2}(T^*\mathbf{X})$ the respective closures. Usual calculus rules hold in these spaces (by density of test forms): we will use e.g. the chain rule. For $\omega, \omega' \in H_C^{1,2}(T^*\mathbf{X})$, one has $\langle \omega, \omega' \rangle \in H^{1,1}(\mathbf{X})$, with

$$\mathrm{d} \langle \omega, \omega' \rangle = \nabla\omega((\omega')^\sharp, \cdot) + \nabla\omega'(\omega^\sharp, \cdot), \quad (2.2)$$

Result

The density statement fails for the covariant Sobolev space $W_C^{1,2}$. The obstruction is already present on the compact interval. Test functions for the metric-measure interval satisfy Neumann boundary conditions, so their differentials vanish at the endpoints. The maximal covariant Sobolev space still contains the parallel 1-form $\mathrm{d}x$, which has nonzero endpoint trace.

Theorem 1. *Let $I = [0, 1]$ be equipped with the Euclidean distance and Lebesgue measure. Then I is an $\text{RCD}(0, \infty)$ space and*

$$\text{TestForm}(I) \quad \text{is not dense in} \quad W_C^{1,2}(T^*I).$$

*More precisely, the $W_C^{1,2}$ -closure of $\text{TestForm}(I)$ is contained in $H_0^1(I) \mathrm{d}x$, while the constant form $\mathrm{d}x$ belongs to $W_C^{1,2}(T^*I)$.*

Interval calculus

We use the standard identification of first-order calculus on the interval:

$$W^{1,2}(I) = H^1(I), \quad L^2(T^*I) = L^2(I) \, dx.$$

Under this identification, a 1-form is written $u \, dx$. The maximal covariant Sobolev space is

$$W_C^{1,2}(T^*I) = H^1(I) \, dx, \quad \nabla(u \, dx) = u' \, dx \otimes dx,$$

and hence

$$\|u \, dx\|_{W_C^{1,2}}^2 = \int_0^1 |u|^2 \, dx + \int_0^1 |u'|^2 \, dx.$$

This is the usual weak covariant derivative on the flat one-dimensional manifold with boundary. It is also what the distributional definition of ∇ gives on this metric-measure model, since test vector fields probe the ordinary weak derivative in the interior and impose no vanishing condition on the form being differentiated.

Lemma 1. *If $f \in D(\Delta)$ for the Cheeger Laplacian on I , then $f \in H^2(0,1)$ and $f'(0) = f'(1) = 0$ in the trace sense.*

Proof. The Cheeger energy on I is

$$\text{Ch}(f) = \frac{1}{2} \int_0^1 |f'|^2 \, dx \quad (f \in H^1(I)).$$

Thus $f \in D(\Delta)$ means that there is $h \in L^2(I)$ such that, up to the source's harmless sign convention for Δ ,

$$\int_0^1 h \varphi \, dx = - \int_0^1 f' \varphi' \, dx \quad \forall \varphi \in H^1(I).$$

Testing first with $\varphi \in C_c^\infty(0,1)$ shows that the distributional second derivative of f belongs to $L^2(0,1)$, so $f \in H^2(0,1)$. For arbitrary $\varphi \in H^1(I)$, integration by parts gives

$$- \int_0^1 f' \varphi' \, dx = \int_0^1 f'' \varphi \, dx - f'(1) \varphi(1) + f'(0) \varphi(0).$$

Comparing with the defining identity and using arbitrary endpoint values of φ yields $f'(0) = f'(1) = 0$. $\square$

Lemma 2. *Every test 1-form on I has zero endpoint trace as an H^1 coefficient. That is, if $\omega \in \text{TestForm}(I)$ and $\omega = u \, dx$, then*

$$u \in H^1(I), \quad u(0) = u(1) = 0.$$

Proof. It is enough to consider one summand $g \, df$, with $f, g \in \text{Test}(I)$. By the previous lemma, $f' \in H^1(I)$ and $f'(0) = f'(1) = 0$. Also $g \in \text{Test}(I)$ satisfies $|dg| \in L^\infty$, so on the interval g has a Lipschitz representative. The coefficient of $g \, df$ is gf' . Since $g \in W^{1,\infty}(I)$ and $f' \in H^1(I)$, the product gf' belongs to $H^1(I)$, and its endpoint traces are

$$(gf')(0) = g(0)f'(0) = 0, \quad (gf')(1) = g(1)f'(1) = 0.$$

Finite sums preserve the same property. $\square$

Proof of the theorem

Let

$$\mathcal{C} := \overline{\text{TestForm}(I)}^{W_C^{1,2}} \subset W_C^{1,2}(T^*I).$$

Writing a form as $u \, dx$, the $W_C^{1,2}$ -norm is exactly the H^1 norm of u . The trace map

$$\text{Tr} : H^1(I) \longrightarrow \mathbb{R}^2, \quad \text{Tr}(u) = (u(0), u(1)),$$

is continuous. Since every test-form coefficient has trace $(0,0)$, every element of the closure $\mathcal{C}$ also has zero trace. Hence

$$\mathcal{C} \subset H_0^1(I) \, dx.$$

Now consider the constant 1-form

$$\omega_0 = dx.$$

Its coefficient is $u \equiv 1$, which belongs to $H^1(I)$, and

$$\nabla \omega_0 = 0.$$

Thus $\omega_0 \in W_C^{1,2}(T^*I)$. But

$$\text{Tr}(1) = (1, 1) \neq (0, 0),$$

so $\omega_0 \notin \mathcal{C}$. Therefore $\text{TestForm}(I)$ is not dense in $W_C^{1,2}(T^*I)$.

Verification notes

The argument uses only the one-dimensional Sobolev trace theorem and the standard identification of the Cheeger calculus on a compact Euclidean interval. The key mechanism is the mismatch between:

$$\text{Test}(I) \subset D(\Delta_{\text{Neu}}) \quad \Rightarrow \quad df = f' dx \text{ has zero endpoint trace,}$$

and

$$W_C^{1,2}(T^*I) = H^1(I) dx,$$

which contains forms with arbitrary endpoint trace.

Limitations and novelty check

This packet addresses the covariant Sobolev space $W_C^{1,2}$. It does not settle the analogous density question for the Hodge Sobolev space $W_H^{1,2}$, where the codifferential may encode a different boundary condition.

The claim should be read with the standard convention, also used in the source passage, that $W_C^{1,2}(T^*X)$ is the maximal space of L^2 1-forms whose distributional covariant derivative lies in L^2 , while $H_C^{1,2}(T^*X)$ is the closure of test forms. If one redefined $W_C^{1,2}$ itself to mean the closure, the density statement would be tautological; that is not the convention indicated by the source.

The bounded novelty check searched the run indexes for arXiv:2308.16294 and for phrases around TestForm , density, $W_C^{1,2}$, $H_C^{1,2}$, covariant Sobolev spaces, and RCD spaces. Web searches for exact phrases around “ TestForm dense” and “ $W_C^{1,2}$ ” found the source paper and related background, but no later public answer or pre-existing interval counterexample in this bounded search.

References

- A. Carbonaro, L. Tamanini, and D. Trevisan, *Boundedness of Riesz transforms on $\mathrm{RCD}(K, \infty)$ spaces*, arXiv:2308.16294, 2023.
- N. Gigli, *Nonsmooth differential geometry – an approach tailored for spaces with Ricci curvature bounded from below*, Mem. Amer. Math. Soc. 251 (2018), no. 1196.